Gouri Lohiya

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.78
Narayana	Class XII (CBSE)	2024	89.20%
St Patrick's Academy	Class X (ICSE)	2022	97.80%

SKILLS

Programming Languages: Python, C++, C, Java

AI & Machine Learning: Machine Learning, Artificial Intelligence, NumPy, Pandas, scikit-learn

Mathematics: Linear Algebra, Probability & Random Variables, Calculus, Differential Equations

Data Structures & Algorithms: Data Structures, Algorithms, Problem Solving

Bioinformatics: Sequence Analysis, BLAST, NCBI, FASTA/FASTQ

Development Tools: Git, GitHub, VS Code

Database: SQL (Basics)

PROJECTS

SMS Spam Classifier | https://github.com/gourilohiya06-arch/SMS-Spam-Classifier

- Context: Developed a Python-based machine learning system to automatically classify SMS messages as spam or legitimate, addressing the need to identify unwanted messages from their textual content.
- Action: Preprocessed SMS text and converted messages into numerical features using TF-IDF, then trained and compared Support Vector Machine and Multinomial Naive Bayes classifiers for spam classification.
- Result: Built a text classification pipeline that evaluates predictions using accuracy, F1-score, and confusion matrix, demonstrating practical application of NLP and supervised machine learning.

Huffman File Compression | https://github.com/gourilohiya06-arch/Huffman-Coding

- Context: Developed a C++-based lossless file compression and decompression system using Huffman Coding to reduce the storage size of text files while preserving data integrity during reconstruction and retrieval.
- Action: Implemented frequency analysis, priority queues, binary trees, and greedy algorithms to construct Huffman trees and generate variable-length binary codes, along with separate encoding and decoding modules for efficient file processing.
- Result: Built a functional compression system capable of converting text files into compressed binary data, storing the required Huffman coding information, and decoding the compressed data back into the original file using tree traversal.

LRU Cache | https://github.com/gourilohiya06-arch/LRU-Cache

- Context: Developed a C++-based Least Recently Used (LRU) Cache to efficiently store and retrieve frequently accessed key-value data while automatically managing limited cache capacity.
- Action: Implemented a combination of hash maps and doubly linked lists to provide fast key lookup, maintain Most Recently Used to Least Recently Used ordering, and dynamically evict the least recently used entries when the cache reaches capacity.
- Result: Built a functional cache supporting O(1) average-time lookup and update operations, with automatic eviction and reordering of cached elements based on usage.

RELEVANT COURSES

Programming Fundamentals, Data Structures & Algorithms, Artificial Intelligence, Algorithms & Computational Problem Solving, Machine Learning, Database Systems & Big Data, Programming in Python & R, Linear Algebra, Probability &

Random Variables, Calculus, Differential Equations

POSITIONS OF RESPONSIBILITY

Finance and Consulting Club: PR Associate

- Coordinated promotional activities and member outreach.
- Assisted in organizing club events and increasing participation.

Elan and nVision Club: Hospitality Associate

- Managed hospitality operations during the fest.
- Coordinated with volunteers and guests to ensure smooth event execution.